Adobe AEM 6.5 & AEM as Cloud Service

Developer (AEM sites) Training
Description – Class room Training
Duration: 45-60 hours

Course Description.

AEM Developer Training teaches developers the fundamentals of building a custom AEM application, based on templates and components. This includes practice with AEM development techniques such as developing templates and authoring environments for mobile sites and websites, dynamic image rendering, navigation and modularization, using CRXDE and any IDE. In-depth lecture presentations and discussions introduce concepts essential to AEM while comprehensive hands-on exercises, using CRXDE and IDEs, reinforce concepts learned during lecture presentations.

This also covers the latest features from AEM as cloud services for developers understanding. Also, a comparative study of features between AEM 6.5. and AEM as cloud service. After we finish, we will discuss on how to prepare for certification.

Objectives

The overall goal of this training is to enable developers to create and understand basic AEM development practices and to prepare themselves for project work under the lead of a senior web developer. This course is not intended to provide any custom instruction focused on specific sites or projects. But it is a launchpad for developers to kickstart their AEM development journey. After this training they should be able to customize any feature by their own and should be able to create and maintain any AEM project.

Target audience

Developer class is designed for developers who wish to learn more about application development with the AEM Platform.

Prerequisites

- Minimum 1 year experience in web development or at least real time experience on any web development project (understanding of project development cycle).
- Basic working knowledge of HTML, CSS.
- Strong knowledge and experience with JavaScript, JAVA.

Software requirement

- A Valid AEM quick start jar or cloud sdk
- A Valid license key (license.properties).
- JDK 8 or JDK 11
- 50 GB free hard disk drive space.
- 8 GB RAM or Higher preferred.
- Apache Maven anything above version 3.5.x and below 3.8.5
- Eclipse or IntelliJ

- AEM Repo tool
- Cygwin for windows

Course content

Basic: (24-28 Hours)

Day 1

Chapter 1: Overview of AEM

- Overview of AEM.
- Pre-requisites of AEM Development
- AEM History from Day CQ to AEM as cloud
- Technology Stack - Basics
- Technology Stack – Java Content Repository
- Technology Stack – Apache Sling
- Technology Stack – Granite Platform
- Demo -: AEM Walkthrough over various consoles
- Demo -: AEM Sites, Assets, Forms, Screens
- AEM Environments – Author, Publish overview
- AEM micro-kernel

Chapter 2: Introduction to AEM as cloud service

- Overview of AEM as a cloud service.Technology Stack - Basics
- On-Premises vs AMS vs Cloud setup
- Cloud internal architecture
- Cloud environments based on run modes
- Everything as service infrastructure
- Development with AEM cloud SDK
- Demo -: Walkthrough of various screens from cloud manager

Chapter 3: Installation

- Prerequisite -: JDK, Apache maven
- Exercise -: Installation of Author and Publish Instances
- Demo -: AEM Folder Structure
- Demo -: Authoring and Publish(Replication) demo
- Modes of AEM - Classic UI vs Touch UI.
- Run Modes
- Exercise on run modes

Chapter 4: Authoring Basics

- WCM Modes.
- Exercise - Creating Pages
- Exercise - Editing Pages
- Exercise - Use of OOTB components
- Replication/Publishing content

- Exercise - AEM Site Creation

Day 2

Chapter 5: Leveraging Developer Tools

- The Web Console
- CRXDE Lite
- Package manager - Working with Packages
- Package share
- IDE (Eclipse, IntelliJ) – brief (details later).
- Exercise – Create and install packages.
- Adobe Id creation.

Chapter 6: Introduction to Content Rendering

- Folder structure of the repository
- Exercise – Create the project structure in /apps
- Page rendering components
- Exercise – Create a page rendering component
- Exercise – Create content to be rendered
- Sling Framework in depth.
- Understanding the sling resolution process
- Basic steps of processing requests
- Decomposing the URL
- Mapping requests to resources
- Locating and rendering scripts
- Exercise – Basic Sling resource resolution and mapping to a rendering script/files
- Exercise – Selector Manipulation

Day 3

Chapter 7: Templates – Only Static template

- What is a template
- Properties of a template
- Exercise – Create a template
- Exercise – Test the contentpage template
- Exercise – Restrict Template use
- Exercise – Add content structure to the Template
- Creating the Website Structure
- Exercise – Create the pages for the site.

Chapter 8: Sightly – Introduction

- Working with sightly

- Sightly syntax
- Page-rendering scripts
- Use APIs to display basic page content
- Modularize the page component
- Exercise – Render basic page content
- Modularize the contentpage component
- Use of HTL Read Eval Print Loop in local.

Chapter 8: Inheritance

- Inheriting components
- Types of hierarchies
- Use of sling:resourceSuperType property
- Exercise – to understand inheritance

Chapter 9: HTL Component and dialog

- Exercise – Creation of a content component (HTL component)
- Exercise – Create edit dialog
- Exercise – Modularize content by using multiple scripts
- Exercise – Create a context-aware configuration

Day 4

Chapter 10: Editable Template

- Responsive layout editing
- Introduction to editable templates
- Exercise - Creating editable templates
- Exercise - Create pages using editable templates
- Understanding editable templates
- Structure vs Initial
- Properties and policies
- Exercise – Create your own custom editable template
- Exercise – Create site structure using custom editable templates

Day 5

Chapter 11: Client-Side Library

- Intro to client-side libraries
- Structure of client-side libraries
- Organizing client-side libraries
- Referencing client-side libraries
- Exercise on client-side libraries

Chapter 12: Core Components

- Structure of a Component
- Core component library
- Client libraries for core components
- Proxy components
- Adding content policies to proxy components
- Developing with core components
- Author with core components

Day 6

Chapter 13: Components Hands On

- Structure of a Component
- Structure components and content components
- Sling Models
- Use JS
- WCMUsePojo
- Exercise – Create a header component
- Exercise – Configuring edit dialog
- Exercise – Add a component client-library to the component
- Exercise – Design dialog

Chapter 14: Custom Components

- Exercise – Create a custom component
- Exercise – Dialog field validation to the custom component
- Exercise – Base style to the custom component
- Exercise – Style system to the custom component
- Exercise – Sling Model as business logic to the custom component
- Various Injectors (1).

Chapter 15: Customizing AEM & AEM with IDEs

- Sling Resource Merger
- Overlays
- Difference between overlay and override
- Exercise – Customizing AEM consoles.
- Eclipse AEM plugin
- IntelliJ with AEM
- Exercise – Customizing AEM tools console.

Advanced: (22-26 Hours)

Day 7

Chapter 16: Coding with Eclipse/IntelliJ

- Exercise - Project creation using latest archetype (capstone my-site project)
- Understand the project structure.
- Importing project into IDE
- What is archetype
- Write business logic into your sling model
- Various type of backend java classes into AEM
- Cygwin installation
- AEM repo installation
- Exercise – Move your code from CRX to IDE and vice versa
- Exercise - Import code developed into CRXDE to mysite project

Day 8

Chapter 17: Sling Model and Sling Model Exporter

- Understanding sling models in details
- Exercise – Write your own Sling Model
- Understanding Sling Model Exporter
- Exercise - Use of sling model exporter
- Exercise – Create your own component with sling model having business logic
- Exercise – Sling Models real time use cases.
- Writing sling models using interfaces and implementation classes
- Sling Delegation framework
- Exercise – sling delegation framework in use.

Day 9

Chapter 18: Services, Components, Servlets

- Software Modularity
- What is OSGi
- OSGI Bundle Life Cycle.
- What is OSGi Service and Component
- Exercise – Create and understand Services and components
- Exercise – Integrating services into sightly and Java
- What is Sling Servlets
- Exercise – Create Resource type based servlet

- Exercise – Create path based servlet
- Exercise – Create post servlet
- Understand various annotations(R7, DS6 etc.)
- Exercise – Use case to handle dynamic requests using servlets
- Exercise – More Servlet use cases.

Day 10

Chapter 19: Logging and Debugging

- Introduction to logging
- Various log levels
- Exercise - Create custom log files
- Exercise - Debugging code in eclipse/Intellij

Chapter 20: User Administration and Security

- Users and Groups in AEM
- Permissions in AEM
- Permissions and ACLs
- Impersonating another User
- Service User in AEM and its uses
- Exercise – Creating Users and Groups in AEM
- Exercise – Creating service users to get resource resolver and session

Day 11

Chapter 21: Workflows

- What is a workflow in AEM
- Workflow models
- AEM content approval workflow
- Exercise – Creating workflow model
- Exercise – Triggering workflow
- Exercise – Take actions on workflows
- Asset Upload workflow with exercises
- Exercise – Create custom DAM update asset workflow
- Exercise – Add a launcher to custom DAM update asset workflow
- Creating custom workflow process step
- Exercise – Use you own workflow process step into custom workflow

Day 12

Chapter 22: Content Fragments(CF) & Experience Fragments(XF)

- What is Content Fragment
- What is Experience Fragment
- What is difference between CF and XF
- Exercise – Create Content fragment model
- Exercise – Create CF using CF model
- Exercise – Create XF template
- Exercise – Create XF using XF template
- Uses of CFs and XFs

Chapter 23: Schedulers and Listeners in AEM

- Schedulers in AEM.
- Overview and configuration of schedulers in AEM.
- Exercise – Sample Scheduler creation.
- Listeners in AEM.
- Overview and configuration of listeners in AEM.
- Exercise – Sample Listeners creation.
- Exercise – Custom schedulers

Day 13

Chapter 24: Indexing and search

- What is Query builder in AEM
- How to use query builder
- How to debug query
- Exercise – Write your own query and use in a component
- Exercise – Develop a component having search functionality utilizing the query

Chapter 25: Multisite Manager

- Multi site management (MSM) overview.
- Exercises on MSM.

Chapter 26: FE Integration and 3rd Party Integration

- AJAX Implementation.
- Sample AEM servlet using AJAX implementation.
- AEM as headless with example OOTB implementation

Assignment -: AEM Custom project from scratch(Demo needs to be given on last day).

Day 14

Chapter 27: Miscellaneous topics

- Asset integration.
- How to deal with migration projects.
- How to deal with Upgrade projects.'
- What to plan for cloud migration
- Deploying code with cloud
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Chapter 28: Dispatcher

- Dispatcher Overview.
- Dispatcher config files understanding

Chapter 29: Edge Delivery Services

- Introduction to EDS
- Fundamental behind EDS evolution
- Architecture and document-based authoring
- Traditional AEM vs EDS
- Crosswalk
- Demo -: Cloud Manager an overview
- Exercise -: Small exercise to understand EDS.

Day 15

Chapter 30: Assessment and Demo

- How to target certification
- Mock test – Online certification exam
- **Demo -: Learning project demo**

Notes -:
1. There will be various quiz (mock tests - multiple choice questions) during the whole training as well as assignment will be provided after each session to assess the knowledge of the candidates.
2. After the training detailed feedback of each candidate will be provided to the management based on the quiz scores and assignment completion status.
3. Total of 3 demo project will be completed during the training.